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Real-time feedback control of penetration stability in al alloy laser welding based on a parallel self-learning fuzzy neural network controller

[Yuan Wang^a](#), [Bangyan Xu^b](#), [Congyi Wu^a](#), [Ping Jiang^a](#), [Chu Han^a](#), [Leshi Shu^a](#)   [Show more](#) ▼[View PDF](#)[Download full issue](#)[Cite](#)[Add to Mendeley](#)[Share](#)[10.1016/j.optlastec.2025.114258](#) ➤[Article](#)[Recommended articles](#)[Cited by \(2\)](#)[Metrics](#)

Abstract

In aluminum alloy laser welding, penetration stability is a key factor affecting weld quality consistency. This study proposes a parallel self-learning control framework based on Fuzzy Neural Network (FNN) to achieve real-time feedback control of penetration stability. Firstly, the high-sensitivity spectrometer is used to monitor the electron temperature characteristics of the plasma in real-time, and the correlation between electron temperature and penetration depth fluctuation is established as the control input. Secondly, a system structure combining the FNN and the PD controller in parallel is designed, and the process database representing prior welding knowledge is embedded

Outline

[Abstract](#)[Keywords](#)[1. Introduction](#)[2. Experimental process and monitoring s...](#)[3. Parallel self-learning fuzzy neural netw...](#)[4. Real-time control of weld penetration s...](#)[5. Conclusions](#)

into the antecedent network of the FNN. Thirdly, the PD controller regulates the process, and the FNN simultaneously conducts online learning using the teacher data provided by the PD controller. Finally, the FNN controller that integrates process knowledge and control dynamics performs feedback control. The welding power is adjusted in real-time based on the monitored electron temperature to stabilize the penetration depth. To validate the proposed method, two typical experimental scenarios were designed: standard flat plate welding to simulate ideal welding conditions and variable-thickness plate welding to simulate the defocus variation conditions in actual processing. Experimental results demonstrate that the proposed real-time control approach effectively stabilizes penetration across different processing conditions. The Coefficient of Variation (CV) of weld penetration depth in flat plate was decreased to 11.08%, while the CV of gradient plate and stepped plate decreased to 10.25% and 12.07%, respectively.

Keywords

Aluminum alloy laser welding; Real-time feedback control; Fuzzy Neural Network; Spectral monitoring

[Previous article in this issue](#)

[Next article in this issue](#)

1. Introduction

Aluminum alloy with low density and high specific strength is widely used in aerospace and automotive applications [[1], [2], [3]]. Laser welding provides high energy density and large weld depth-to-width ratio, serving as a key process for aluminum alloy joining [4,5]. However, aluminum alloy laser welding is often highly unstable due to the low density and viscosity of the molten metal, leading to severe penetration depth fluctuations and frequent inconsistencies in weld quality [6,7]. Traditional pre-welding parameter optimization methods are inadequate for handling the transient instability

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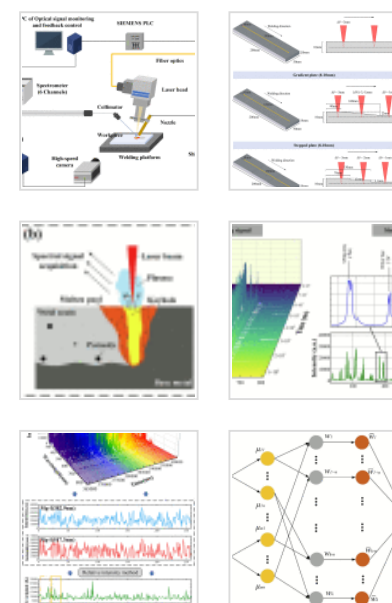
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[References](#)

[Show full outline](#) 

Figures (18)



[Show 12 more figures](#) 

Tables (8)

 Table 1. Chemical compositions of 6061 ...

 Table 2. Process conditions used in the e...

fluctuations in aluminum alloy laser welding processes. Therefore, there is an urgent need for a method to achieve feedback control of process parameters based on dynamic weld penetration variations during welding, ensuring consistent weld quality.

The prerequisite for feedback control in aluminum alloy laser welding is obtaining suitable online monitoring signals that accurately reflect welding quality [8]. Several typical monitoring devices include Optical Coherence Tomography (OCT), vision sensors, and spectrometers. OCT enables direct keyhole monitoring and 3D reconstruction [9]. Deyuan Ma et al. [10,11] used OCT to capture keyhole depth fluctuations, enabling the detection of welding defects. However, the system experiences significantly increased latency during massive interferometric signal acquisition and Fourier transform processing. Additionally, its high sensitivity to welding environments (e.g., spatter) and expensive equipment costs make it challenging for industrial deployment as a real-time monitoring solution for feedback control. Vision sensors capture molten pool morphology to monitor keyhole dynamics and spatter behavior. Wang Cai et al. [12,13] proposed a strategy using two deep learning neural networks combined with image processing to diagnose penetration states, achieving highly robust real-time penetration depth predictions with an accuracy of 98.68%. Nevertheless, conventional industrial cameras can only capture the 400–700nm visible spectrum, and require 3D convolutional networks for molten pool edge detection[14,15], resulting in significant latency issues[16]. Spectrometers monitor plasma emissions to evaluate welding process status. Jiyoung Yu et al. [17] collected spectral signals from the plasma plume in the 225–975nm range during laser welding and achieved 90% accuracy in penetration state monitoring. Shixuan Li et al. [18,19] proposed a high-precision penetration depth monitoring method for Al-Cu laser lap welding based on spectral signals, achieving a monitoring accuracy of 0.05mm. Spectroscopic signal monitoring offers broader spectral coverage and multi-dimensional information acquisition. The combination with more efficient data processing techniques makes it more promising for real-time adaptability in laser welding feedback control.

Automatic Process Control (APC) technology enables dynamic feedback compensation to continuously correct deviations, significantly improving manufacturing precision and

- Table 3. Parameters of Mg-I Atomic Line.
- Table 4. Parameters for WMA Calculation.
- Table 5. Processing parameters of proce...
- Table 6. Reference electronic temperatu...

[Show 2 more tables](#) ▼

efficiency [20]. In real-time laser welding control, traditional controllers include expert systems, PID, and fuzzy logic [[21], [22], [23], [24]]. M. Seibold et al. [25] developed an expert system based on ADWin-Pro II to regulate penetration depth and control the bonding time of aluminum-copper lap joints. Teresa Sibillano et al. [26] used plasma electron temperature as a process-monitoring signal and adjusted laser power via a PD feedback controller to minimize temperature variations and maintain the desired penetration depth. Traditional control systems often rely on linearity assumptions and manual parameter tuning. When welding disturbances such as keyhole collapse or spatter surges arise, their conservative design strategies that compromise performance and remain inadequate for adapting to complex control environments.

Data-driven machine learning algorithms can automatically establish high-order nonlinear mappings and continuously optimize models based on data dependencies [27]. In real-time laser welding feedback control tasks, supervised learning and reinforcement learning have shown strong applicability. The former builds precise mappings between process parameters and quality indicators using labeled data [28], while the latter autonomously optimizes multi-objective control strategies through environmental interactions [29,30]. Both approaches address the limitations of traditional methods in nonlinear modeling and dynamic adjustments [31]. Qi Zhong et al. [32] proposed a feedback model based on a multi-input neural network, transforming discrete sintering states in laser processing into material quality trends to ensure ideal thermal variations during sintering. Mohammad et al. [33] used CCD cameras to monitor laser processing layer height and designed a neuro-fuzzy model controller to regulate travel speed for uniform layer height. Johannes Günther et al. [34] developed a laser welding control system integrating deep learning features with actor-critic reinforcement learning and validated the learning strategies through simulations. Giulio Masinelli et al. [35] applied both Q-learning and policy gradient reinforcement learning for penetration depth regulation, introducing a new approach to process parameter feedback control.

However, existing research on real-time feedback control of laser welding parameters still faces methodological limitations. The supervised learning paradigm heavily relies on

expert-labeled data, and its training outcome is often constrained by local optima or empirical compromises [36]. Meanwhile, the reinforcement learning paradigm requires millions of training steps to converge and demands strict constraints on policy updates to prevent process instability [37]. It is worth noting that few current studies explore the integration of prior process knowledge. Structuring and embedding process knowledge can effectively reduce the policy search space and improve learning efficiency, while also enhancing control accuracy through physical mechanism guidance. There is an urgent need to develop a data-driven architecture integrated with prior knowledge, which incorporates laser welding physical mechanisms to achieve rapid control strategy learning and stable real-time control.

Moreover, most existing laser welding real-time feedback control research remains at the laboratory stage, with results primarily based on simulation or simplified welding tests under ideal conditions. However, in actual welding processes, defocus variation caused by workpiece thermal deformation and laser head thermal lensing effects significantly compromises weld consistency. Regulating such welding quality fluctuations induced by actual operating conditions remains a critical challenge for industrial applications [38].

This study innovatively proposes a parallel self-learning Fuzzy Neural Network (FNN) controller, which for the first time achieves engineering level real-time feedback control of penetration depth fluctuations in aluminum alloy laser welding. The system acquires plasma electron temperature characteristics through spectral monitoring as core feedback signals correlated with penetration variations. The controller architecture parallelizes a traditional PD controller with the FNN, embedding process knowledge via fuzzy rule-based databases in the antecedent structure of FNN. Using the PD controller's dynamic response outputs as teacher signals for FNN learning, the final FNN controller integrates both process knowledge and control dynamics for real-time feedback. To comprehensively verify the control performance, multiple experimental scenarios were designed: standard flat plate welding was used to simulate ideal conditions, while variable-thickness plate welding simulated dynamically fluctuating defocus conditions in

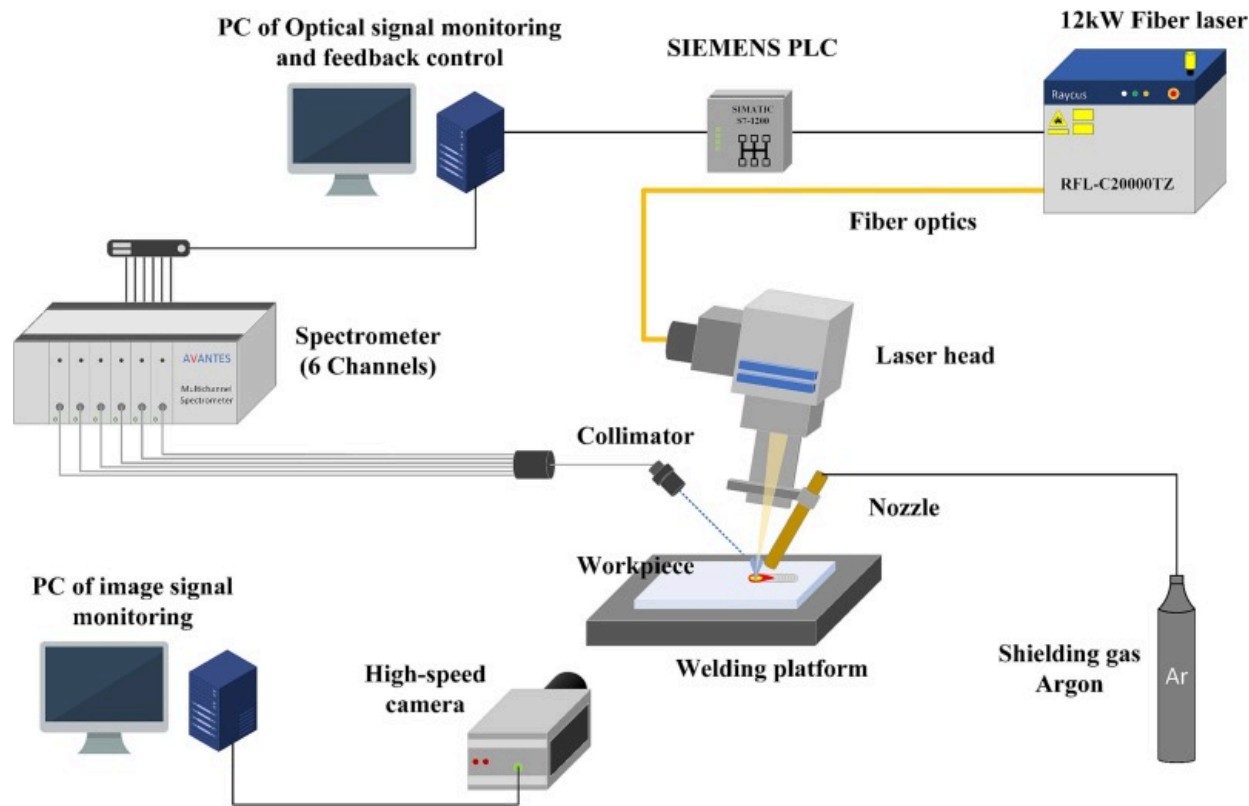
actual processing. The proposed controller successfully achieves real-time feedback control of penetration depth fluctuations under these two representative conditions.

The remainder of this paper is divided into four sections. [Section 2](#) introduces the real-time feedback control platform and the electron temperature signal used as the control input. [Section 3](#) details the FNN control method and the overall architecture of the parallel self-learning controller. [Section 4](#) describes the deployment of the process knowledge database, controller implementation, and validation results. Finally, [Section 5](#) summarizes the main conclusions.

2. Experimental process and monitoring signal

2.1. Experiment process

The complete experimental system includes welding processing module, monitoring module and control module, as shown in [Fig. 1](#). The Raycus RFL-C20000TZ high-power fiber laser with a maximum power of 12kW and an output fiber core diameter of 200 μ m was used as the laser source. Welding was performed using self-developed laser head with single-point gaussian beam mode, spot size of 0.4mm, and focal length of 300mm. Laser head was mounted on KUKA KR70 R2100 six-axis robotic arm for welding motion control. Monitoring module employed AvaSpec-UL32048CI-6-EVO spectrometer to collect optical signals during the welding process. This spectrometer has six channels, enabling precise acquisition of optical signals in the wavelength range of 200nm to 1100nm, with minimum channel resolution of 0.1 nm. Self-developed long-focus spectral acquisition head was aligned with plasma plume above the keyhole, and optical signals were transmitted to the spectrometer through a 1-to-6 fiber bundle, ensuring accurate signal acquisition. A paraxial high-speed camera (Phantom v611) was used to capture plasma dynamics during welding, providing validation for the collected spectral signals and welding process stability. Control module utilized SIEMENS SIMATIC S7-1200 PLC (model CPU 1212C DC/DC/RLY) to transmit control signals from the PC to the laser, achieving real-time modulation of welding process parameters.



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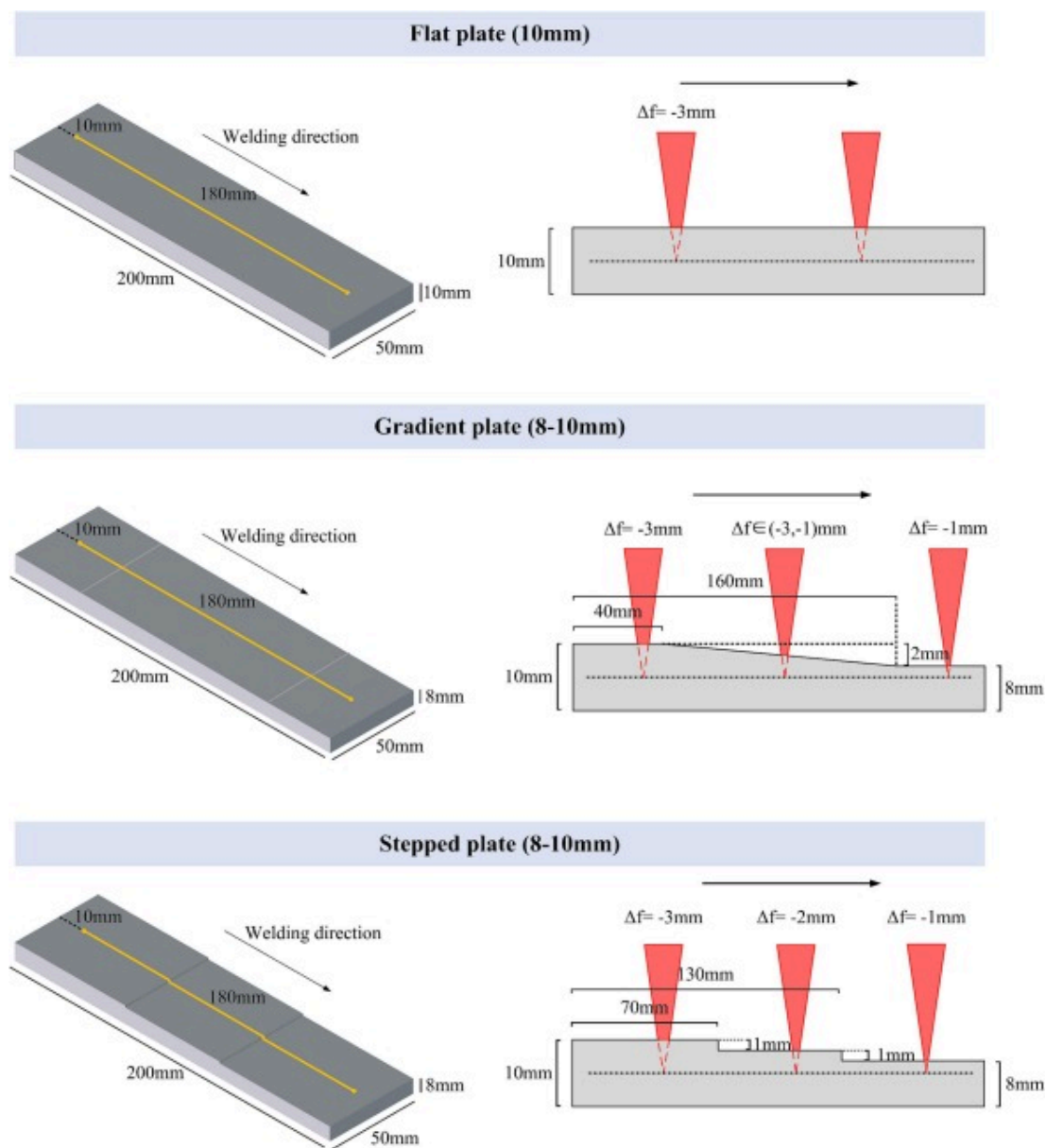
Fig. 1. Experimental platform for real-time monitoring and feedback control of aluminum alloy laser welding.

The workpiece material is 6061 aluminum alloy, which is widely used in laser processing. This Al-Mg-Si alloy offers high strength and toughness, with chemical composition shown in Table 1. Due to its thermal conductivity and viscosity, weld penetration fluctuations and porosity defects are common during welding. Workpieces include 10mm thick flat plate and 8–10mm thick variable-thickness plate, with dimensions and welding positions shown in Fig. 2, where the yellow line indicates the weld seam location. Flat plate simulate stable and ideal processing conditions, while variable-thickness plate simulate

unstable welding conditions caused by workpiece deformation and thickness variations. Variable-thickness plate include gradient plate and stepped plate to test controller performance under continuous and sudden disturbances, respectively. The control objective of this study is to ensure penetration consistency across all thickness regions of the same workpiece, thereby achieving stable and reliable welding quality. In this context, penetration is defined as the vertical distance between the deepest point of the molten zone at any location of the weld and the surface of the workpiece at that location.

Table 1. Chemical compositions of 6061 Al alloy.

Element (%)	Si	Fe	Cu	Mn	Mg
6061 Al	0.4–0.8	≤0.7	0.15–0.4	≤0.15	0.8–1.2
Element (%)	Cr	Zn	Na	Ti	Al
6061 Al	0.04–0.35	≤0.25	-	≤0.15	Bal.



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Fig. 2. Dimensions of workpiece and welding positions.

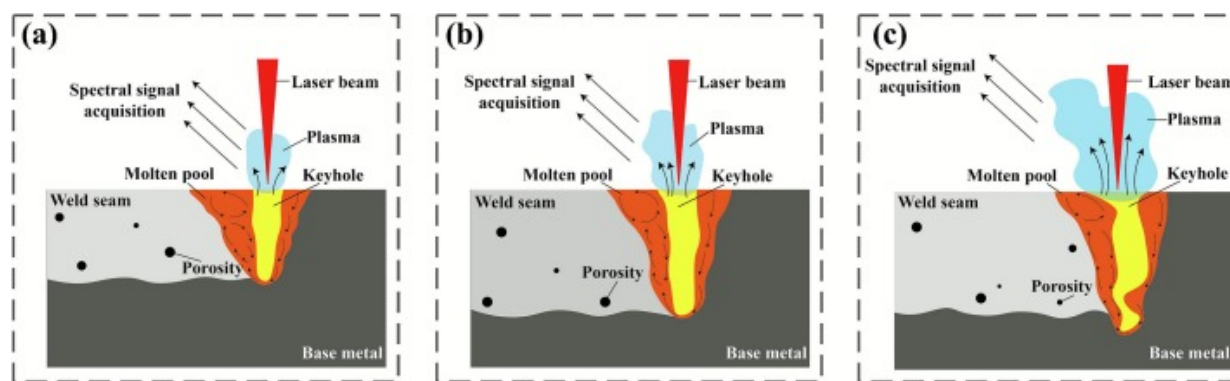
Welding parameters are shown in Table 2, where both laser power and defocus amount represent the initial values at the start of the processing preparation. Experiment #1 involves 10mm thick flat plate, while Experiment #2 involves 8–10mm thick gradient plate and stepped plate. A 5° deflection angle was set in all experiments to prevent equipment damage from high reflectivity during aluminum alloy laser welding. Shielding gas is Ar at a flow rate of 15L/min. The parameters were determined through preliminary experiments to ensure sound weld formation. Workpiece surfaces were polished and cleaned with acetone before welding to remove oxide films that could cause metallurgical pores. Long-focus spectral acquisition head was installed 45° above and 20mm away from the welding point, aligned with the plasma above the keyhole to collect spectral signals in real time. The high-speed camera was positioned 2m from the welding point, capturing plasma dynamics horizontally throughout the entire welding process to analyze welding stability. Experiments #1 and #2 employ two different controller tailored to distinct control objectives. Experiment #1 focusing on flat plate welding, aims to regulate transient penetration depth fluctuations caused by the welding process under stable conditions; Experiment #2 focusing on variable-thickness plate welding, is designed to control macroscopic penetration depth variations under unstable processing conditions.

Table 2. Process conditions used in the experiments.

Experiment No.	Initial laser power	Welding speed	Initial defocusing amount	Deflection angle	Workpiece
#1	6600W	40mm/s	−3mm	+5°	flat plate (10mm)
#2	5600W	40mm/s	−3mm	+5°	gradient plate (8–10mm)/stepped plate (8–10mm)

2.2. Real-time spectral monitoring

Plasma is a crucial signal for monitoring the stability of the welding process, as illustrated in Fig. 3. In Fig. 3(a), under low-power stable welding conditions, Marangoni convection inside the molten pool exhibits uniform flow, contributing to pool stability. The plasma above the keyhole is evenly ejected with relatively low concentration. In Fig. 3(b), under high-power stable welding conditions, the plasma concentration increases compared to the low-power state, but the ejection remains stable. In Fig. 3(c), when severe penetration depth fluctuations occur due to intense Mg evaporation or high-speed molten metal flow, Marangoni convection becomes irregular, further destabilizing the molten pool. Simultaneously, plasma ejection intensifies significantly, and its concentration surges sharply within a short period.



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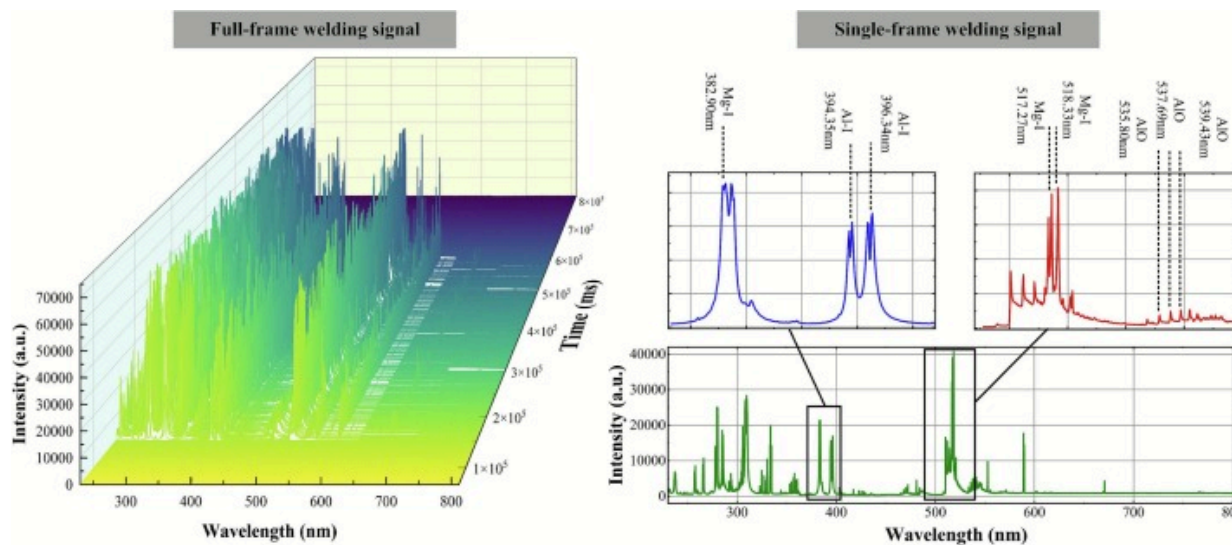
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Fig. 3. Plasma and molten pool under different welding conditions. (a) Stable laser welding plasma and molten pool under low power; (b) Stable laser welding plasma and molten pool under high power; (c) Unstable laser welding plasma and molten pool under severe penetration depth fluctuations.

Spectral intensity/counts represent the intensity of electromagnetic radiation. The counts displayed on the Avasoft software interface (the spectrometer data processing software) correspond to voltage data generated by CMOS pixels in the spectrometer when interacting with electromagnetic radiation [19]. Higher electromagnetic radiation

intensity results in higher voltage output from the CMOS. Therefore, monitoring fluctuations in electromagnetic radiation intensity via the spectrometer enables observation and analysis of plasma optical signals, providing an accurate reflection of welding process stability.

A representative raw spectral signal collected in this study is shown in Fig. 4, with acquisition parameters set as an integration time of 0.5ms and an averaging frame count of 6. The Full-frame welding signal captures the time-dependent plasma characteristic spectral lines of aluminum alloy as it absorbs laser energy. The plasma initiation and termination moments are clearly visible, while the remaining signals correspond to background spectra. Single-frame welding signal intuitively reveals the characteristic spectral lines of plasma generated during aluminum alloy laser welding. These include multiple distinct Al-I atomic lines, Mg-I atomic lines, and band-shaped AlO radical molecular lines. The intensity of these characteristic spectral lines indicates the magnitude of welding dynamics.



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Fig. 4. Plasma characteristic spectral lines of 6061 aluminum alloy laser welding.

The state of plasma is determined by interactions such as excitation, ionization, recombination of internal particles, particle collisions, and photon absorption.

Thermodynamic equilibrium assumes that the interactions among internal particles can be neglected, allowing the plasma state to be directly characterized by macroscopic quantities such as temperature and concentration. In actual laser welding, electron-particle collisions dominate, allowing the plasma to be approximated as being in a local thermodynamic equilibrium (LTE) state. The criterion formula of LTE can be expressed as:

$$ne \geq 1.6 \times 10^{12} (E_2 - E_1)^3 T_e^{1/2} (cm^{-3}) \quad (1)$$

where ne is the electron density, E_2 is the first excited state energy, E_1 is the ground state energy, T_e and is the electron temperature.

The critical electron density generally does not exceed $10^{16} cm^{-3}$. Under normal conditions, the electron density of laser welding plasma ranges from $10^{16} cm^{-3}$ to $10^{17} cm^{-3}$, which exceeds the critical electron density and satisfies the condition for local thermodynamic equilibrium.

When the laser welding system is in LTE, electron temperature and electron density are two key parameters for evaluating plasma state. Electron temperature represents the average kinetic energy of free electrons in the plasma, directly reflecting the welding energy input and the thermal state of the plasma. Furthermore, electron temperature is typically obtained from the spectral intensity ratio method or line-pair intensity ratio, which does not require very high spectral resolution and provides more stable measurements. In contrast, electron density is usually determined from spectral line broadening, which imposes stricter requirements on spectral resolution and signal-to-noise ratio. Numerous studies have demonstrated a distinct correlation between plasma electron temperature and penetration depth [18,39]. Therefore, this study takes electron temperature T_e as the control target to reduce the penetration fluctuation by stabilizing electron temperature.

This study calculates the electron temperature of the plasma using the relative intensity method. It is determined based on the relative intensity of two characteristic spectral lines acquired by the spectrometer, as expressed by the following [40]:

$$Te = \frac{5040(E_1 - E_2)}{\lg \frac{A_1 g_1}{A_2 g_2} - \lg \frac{I_1}{I_2} - \lg \frac{\lambda_1}{\lambda_2}} \quad (2)$$

where A represents the transition probability, g is the statistical weight of the state, E denotes the energy level, and λ is the wavelength of the spectral line. The values of A , g , and E can be obtained from the National Institute of Standards and Technology (NIST) database. When selecting spectral lines, the following criteria should be considered: maximizing the energy level difference between the two lines, minimizing measurement errors in transition probability, and ensuring the absence of self-absorption [41].

In Fig. 4, the spectral intensity of Mg-I atoms is significantly higher than that of Al-I atoms and AlO molecules. This is because the boiling point of magnesium (1380K) is much lower than that of aluminum (2600K), and the excitation of Mg atoms can absorb lower-energy photons from the laser. Moreover, the intense ejection of magnesium during welding is a key factor contributing to welding instability. Therefore, this study selects two Mg-I atomic spectral lines to calculate the electron temperature during welding, with their parameters listed in Table 3.

Table 3. Parameters of Mg-I Atomic Line.

Atomic line	Acquisition wavelength	$A_{ki} (s^{-1})$	$E_i (cm^{-1})$	$E_k (cm^{-1})$	g_k
Mg-I	382.9001	8.99e+07	21850.405	47957.058	3
Mg-I	517.2669	3.37e+07	21870.464	41197.403	3

In actual monitoring, the intense ejection of magnesium and real-time plasma fluctuations introduce instability, resulting in high-frequency oscillations in the raw Te

signal. Excessive fluctuations in the feedback signal can cause frequent control adjustments, exacerbating system overshoot and undershoot, which in turn leads to increased instability in welding due to frequent power adjustments. Therefore, this study employs a sliding window-based low-pass filtering method during monitoring to eliminate high-frequency noise while preserving the primary variation trend of the Te signal, ensuring high real-time performance and stability of the signal processing.

Based on the real-time characteristics of the control system, the monitoring signal must accurately and promptly reflect the current plasma variations. Therefore, a Weighted Moving Average (WMA) filtering method is applied to increase the weight of the current signal. The calculation method can be expressed as:

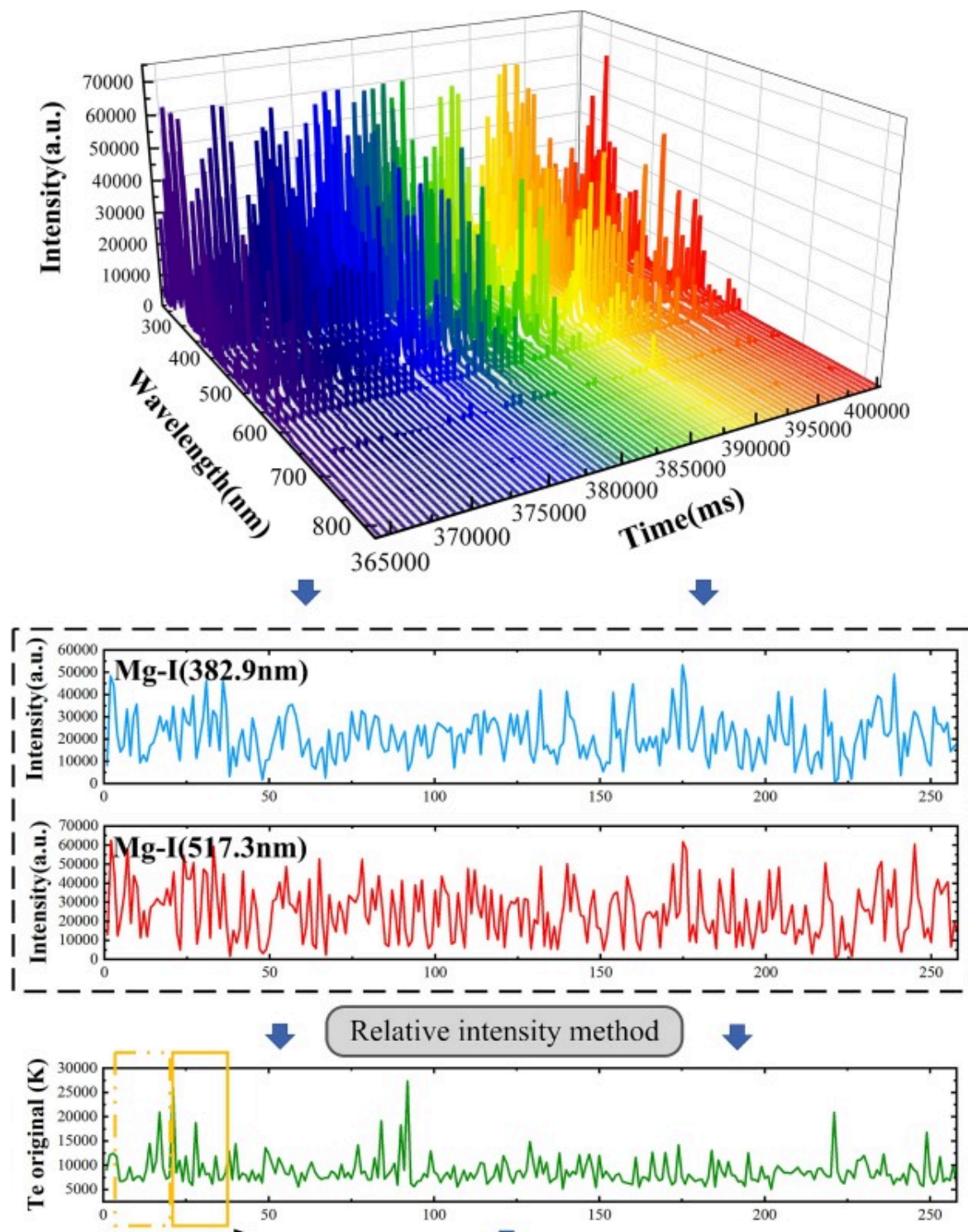
$$Te(t) = \frac{\sum_{i=0}^{N-1} w_i Te_{origin}(t-i)}{\sum_{i=0}^{N-1} w_i} \quad (3)$$

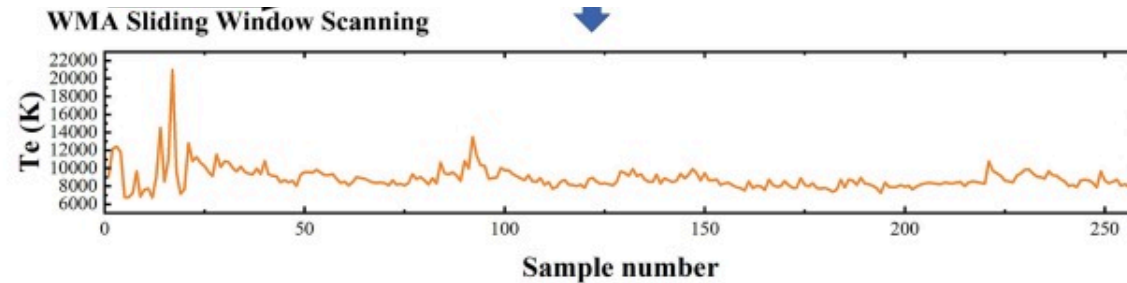
where Te represents the smoothed electron temperature signal, while Te_{origin} denotes the raw electron temperature signal. N is the size of the sliding window, and w is the weight, which increases for time points closer to the current moment.

Since the control objectives for flat plate and variable-thickness plate welding differ, two data processing modes are adopted to accommodate the monitoring requirements of plasma characteristics on different time scales. The flat plate welding mode focuses on capturing short-term intense plasma fluctuations, utilizing a smaller sliding window to enhance sensitivity to transient changes; the variable-thickness plate welding mode is designed to analyze macroscopic plasma evolution trends, employing a larger window to smooth local disturbances and highlight overall variation patterns. The final sliding window parameters, determined through experimental validation, are presented in [Table 4](#). The complete signal processing procedure is shown in [Fig. 5](#).

Table 4. Parameters for WMA Calculation.

Experiment No.	N	ω
#1	10	$\omega_1 = 0.3, \omega_2 = 0.2, \omega_{3-10} = 0.0625$
#2	20	$\omega_1 = 0.2, \omega_2 = 0.1, \omega_{5-20} = 0.03125$





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Fig. 5. Spectral signal processing workflow.

To enable real-time acquisition of plasma spectral signals in the feedback control system, a custom real-time communication module for the AvaSpec-UL32048CI-6-EVO spectrometer was jointly developed with the Avantes distributor. The module utilizes a shared memory mapping mechanism based on the API function `CreateFileMapping`, allowing acquired data to be transmitted in real-time to a designated address in computer memory. The data is organized and stored in blocks, and the Python `mmap` library is used to map and access the shared memory, enabling real-time extraction and parsing of the written data. The developed real-time communication module transmits measurement frame count, timestamps, ten preset acquisition wavelengths, and their corresponding intensity values. The spectral intensity data acquisition program is implemented in Python 3.9. In actual welding processes, the communication time for single-frame spectral signal acquisition is controlled within 1 ms.

3. Parallel self-learning fuzzy neural network controller

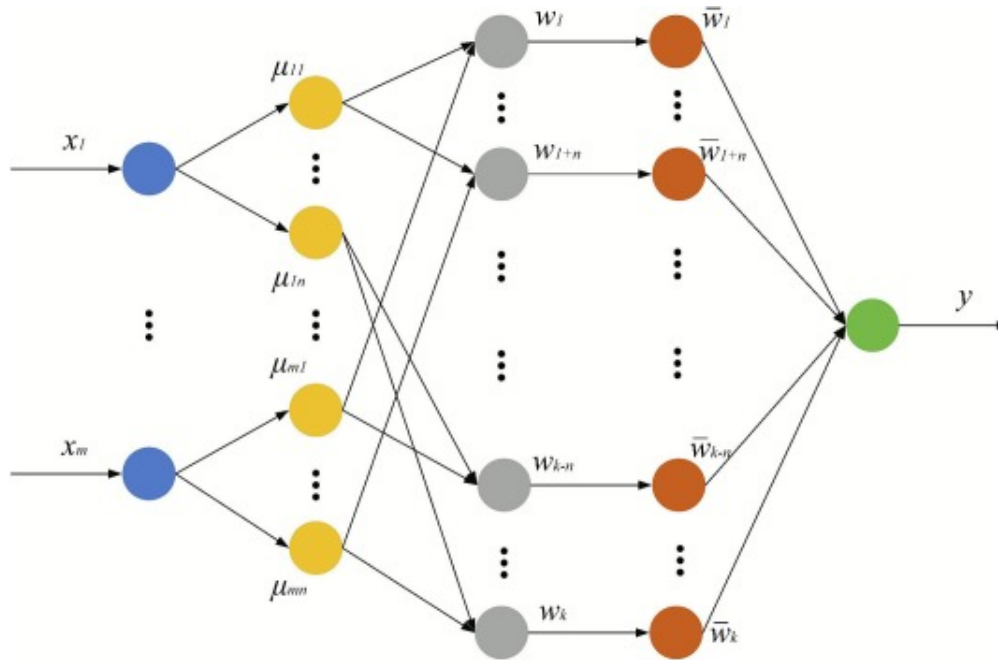
3.1. Fuzzy neural network control method

Aluminum alloy laser welding is a highly transient and nonlinear complex physical system, making it challenging to establish an accurate mathematical model of the

controlled object. Traditional control methods struggle to meet the control quality requirements of complex systems in modern engineering applications. This study integrates a FNN as the primary controller to achieve precise real-time regulation of aluminum alloy laser welding parameters by combining conventional fuzzy systems with neural networks.

The essence of FNN lies in utilizing a neural network to implement fuzzy logic inference, thereby combining the interpretability of fuzzy logic with the learning capability of neural networks [42]. In FNN, fuzzy logic—based on conventional fuzzy control theory—handles uncertainties and nonlinearities, while the neural network adaptively adjusts fuzzy rules and membership functions [43]. This enables the system to continuously optimize the control strategy based on data, thereby improving control precision.

The fundamental framework of FNN consists of five layers, resembling a feedforward neural network, as illustrated in Fig. 6. In FNN, fuzzy rules are expressed in the form of “IF-THEN” statements to describe the fuzzy relationship between input and output variables [44]. The “IF” part of the fuzzy rules serves as the antecedent network, defining the fuzzy conditions of input variables, and is composed of the first four layers. The “THEN” part represents the fuzzy consequences of output variables and is constructed by the fifth layer.



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Fig. 6. Basic network structure of FNN.

The first layer of FNN is the input layer, which directly receives the controller input signals and transmits them to the next layer without performing any calculations. The input value is $\mathbf{x} = [x_1, x_2, \dots, x_m]^T$, and the number of nodes in this layer is m .

The second layer of the FNN is the fuzzification layer, which maps input variables to fuzzy membership functions and calculates their membership degrees in different fuzzy sets. Each node in this layer corresponds to a fuzzy subset in traditional fuzzy logic, such as **NB** and **NP**, representing the membership degree μ_{ij} ($i = 1, \dots, m; j = 1, \dots, n$) of the input variable in each subset. n denotes the number of fuzzy subsets for each input x , and the total number of nodes in this layer is mn . The membership functions are typically Gaussian Membership Function or Triangular Membership Function. If a Gaussian

membership function is used, it is expressed as:

$$\mu_{ij}(x) = \exp\left(-\frac{(x_i - c_{ij})^2}{2\sigma_{ij}^2}\right) \quad (4)$$

where c_{ij} represents the center of the membership function, and σ_{ij} represents the width of the membership function.

The third layer of FNN is the rule-matching layer, which calculates the activation strength of fuzzy rules. Each neuron corresponds to a fuzzy rule (IF-THEN), and the total number of nodes in this layer is k , representing the total number of fuzzy rules under all IF conditions. The product and minimum methods are commonly used to compute rule matching degree, with the former being more suitable for differentiable optimization. When adopting the product method, the calculation is given by:

$$w_i = \prod_{j=1}^n \mu_{ij}(x_j) \quad (5)$$

where w_i represents the activation degree of the i -th rule.

The fourth layer is the normalization layer, which normalizes the activation strengths of all the rules to ensure that the contribution of each rule is reasonable. The total number of nodes in this layer is k . The mathematical expression for normalization is:

$$\bar{w}_i = \frac{w_i}{\sum_{i=1}^k w_i} \quad (6)$$

The fifth layer is the output layer, which is responsible for calculating the final output, i.e., defuzzification. The weighted average method is commonly used for this calculation, the calculation is given by:

$$y = \sum_{i=1}^k z_i \bar{w}_i \quad (7)$$

where z_i represents the consequent, which is the connection weight of the last layer.

In this study, the FNN model also trains and learns parameters through the concept of error backpropagation, primarily involving the optimization of the connection weights in the fifth layer (consequent parameters) and the membership degree center and width in the second layer (antecedent parameters). The target output is defined as y^* , and the mean squared error (MSE) is used as the loss function:

$$E = \frac{1}{2}(y^* - y)^2 \quad (8)$$

The gradients are then computed through backpropagation to optimize the parameters.

The fifth layer performs a weighted average operation. For the rule consequent parameter z_i , it can be optimized through gradient descent. The partial derivative of the error E with respect to z_i is given by:

$$\frac{\partial E}{\partial z_i} = \frac{\partial E}{\partial y} \cdot \frac{\partial y}{\partial z_i} \quad (9)$$

Therefore:

$$\frac{\partial E}{\partial z_i} = -(y^* - y) \frac{w_i}{\sum_{i=1}^k w_i} \quad (10)$$

The parameter z_i can be updated using gradient descent as follows:

$$z_k^{(t+1)} = z_k^{(t)} - \eta \frac{\partial E}{\partial z_k} \quad (11)$$

where η is the learning rate.

In the second layer, the gradient-based optimization algorithm adjusts the center c_{ij} and width σ_{ij} of membership function, with update rules derived from chain-rule partial derivatives of the error function.

The partial derivative of the error with respect to c_{ij} and σ_{ij} are calculated as:

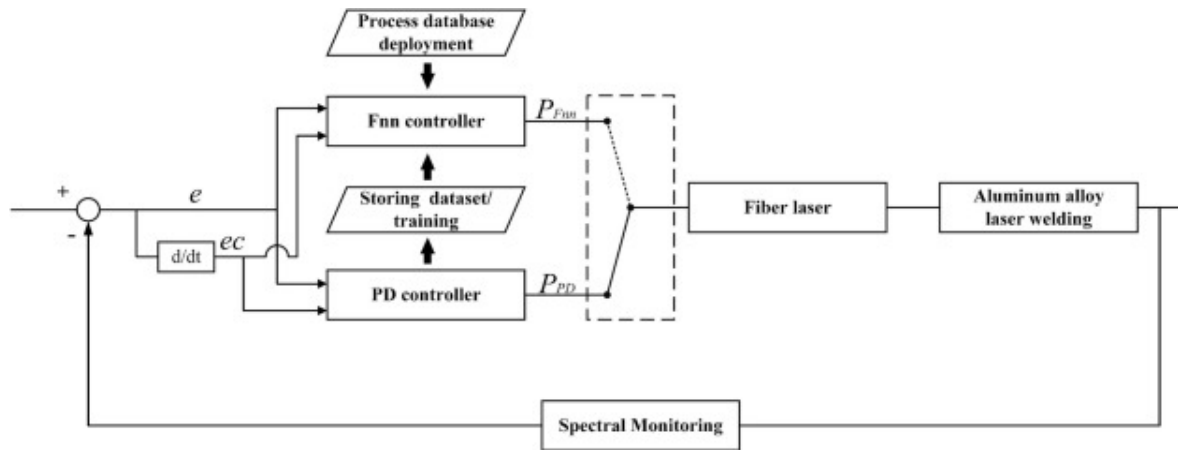
$$\frac{\partial E}{\partial c_{ij}} = \frac{\partial E}{\partial y} \cdot \frac{\partial y}{\partial w_i} \cdot \frac{\partial w_i}{\partial \mu_{ij}} \cdot \frac{\partial \mu_{ij}}{\partial c_{ij}} \quad (12)$$

$$\frac{\partial E}{\partial \sigma_{ij}} = \frac{\partial E}{\partial y} \cdot \frac{\partial y}{\partial w_i} \cdot \frac{\partial w_i}{\partial \mu_{ij}} \cdot \frac{\partial \mu_{ij}}{\partial \sigma_{ij}} \quad (13)$$

3.2. Parallel self-learning control architecture

Directly setting up FNN controller based on expert knowledge rules allows parameter adjustments without requiring training data. However, this approach heavily relies on experience and has limited control effectiveness. In the complex and dynamically changing laser welding process, it is crucial to have high-quality datasets containing key feature information. By training and learning from these datasets, the FNN model achieves adaptive and high-precision control performance.

This study proposes a parallel self-learning FNN control framework, in which the FNN controller operates alongside a conventional PD controller. The function of the PD controller is to provide a learning reference for the FNN controller, generating reference data during the welding process that reflects the dynamic characteristics of the control system. Their inputs are the error e and error change rate ec of the electron temperature relative to the reference value. e is calculated as the standard electron temperature $T_{e_{stan}}$ minus the current electron temperature T_e . ec takes the derivative of e with respect to time. The output P_{Fnn} , P_{PD} are the modulated laser power. The complete control structure is shown in Fig. 7.



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Fig. 7. Parallel self-learning FNN control framework.

The deployment and workflow of the control system are as follows:

- (1) Model Deployment Stage: In the initial setup phase of the FNN, a process database is embedded into the fuzzification layer to establish the initial distribution of membership functions. This process enables the FNN to incorporate prior process knowledge for regulation while preventing overfitting to extensive PD controller experimental data during training;
- (2) Initial Control Stage: During this phase, a pre-tuned and optimized PD controller is employed for feedback control to ensure stable system operation. At this stage, the control signal is directly computed by the PD controller and applied to the controlled object to regulate the target variable;
- (3) FNN Learning Stage: While the PD controller performs feedback control, the system's input and corresponding output are continuously collected to construct a training dataset in real-time. This dataset is used to train and optimize the FNN, allowing it to learn the dynamic characteristics of the welding system;

(4) FNN Control Stage: Once the FNN is trained with a dataset containing comprehensive feature information and its control performance is optimized, the system switches to the FNN, which takes over feedback control independently. At this stage the FNN integrates prior process knowledge and system dynamics, leveraging its nonlinear mapping capability to accurately regulate process parameters.

Since Experiment #1 and Experiment #2 correspond to stable and unstable processing conditions, respectively, their power modulation modes differ.

Experiment #1 simulates the laser welding process under stable conditions. Under ideal conditions, welding with the standard power P_{stan} results in a stable electron temperature Te , so a fixed-reference power modulation mode is adopted. This method is suitable for scenarios where the target power is known under stable conditions. The standard power P_{stan} remains unchanged, and the final power output is obtained by adding the error correction value P_k calculated by the controller to the reference value. The formula can be expressed as:

$$P_k = P_{stan} + \Delta P_k \quad (14)$$

where P_{stan} is the fixed standard power, ΔP_k is the error correction value obtained by the controller at time step k , and P_k is the final power output.

Experiment #2 simulates the laser welding process under varying conditions, where the standard power P_{stan} required to stabilize the electron temperature Te changes in real time. In this case, using the initial power P_{init} as a fixed reference becomes invalid, so a cumulative update power modulation mode is adopted. This approach is suitable for dynamically changing systems, where each adjustment influences subsequent control outputs, aligning with an adaptive regulation framework. The initial reference power P_{init} is used only for initialization. In each step, the modulation value ΔP_k calculated by the controller is directly accumulated to the previous power output, forming a dynamically

changing reference power. The formula can be expressed as:

$$P_k = P_{k-1} + \Delta P_k \quad (15)$$

where P_{k-1} is the power output at the previous time step, ΔP_k is the power adjustment value calculated by the controller at the current step, and P_k is the final power output, which also serves as the new reference value.

The controller designed in this study was developed using Python 3.9 and runs on a Windows 11 environment. The computing hardware includes an NVIDIA GeForce RTX 4060 GPU with support for CUDA 12.5. Test results indicate the power computation time of the FNN ranges from 30 to 35ms, with complete feedback control latency maintained below 40ms.

4. Real-time control of weld penetration stability

4.1. Fuzzy layer deployment based on process database

During the initialization phase of the FNN controller, the process database is embedded into the fuzzification layer to incorporate prior process knowledge and optimize the setup of fuzzy membership functions. Specifically, the distribution characteristics of the error e and the rate of error change ec in the electron temperature Te signal are obtained through experiments to determine the center c and width σ of the fuzzy membership functions. The constructed process database is derived from experimentally validated process parameters, which ensure consistent weld penetration under different plate thickness conditions. The specific selection scheme is shown in [Table 5](#).

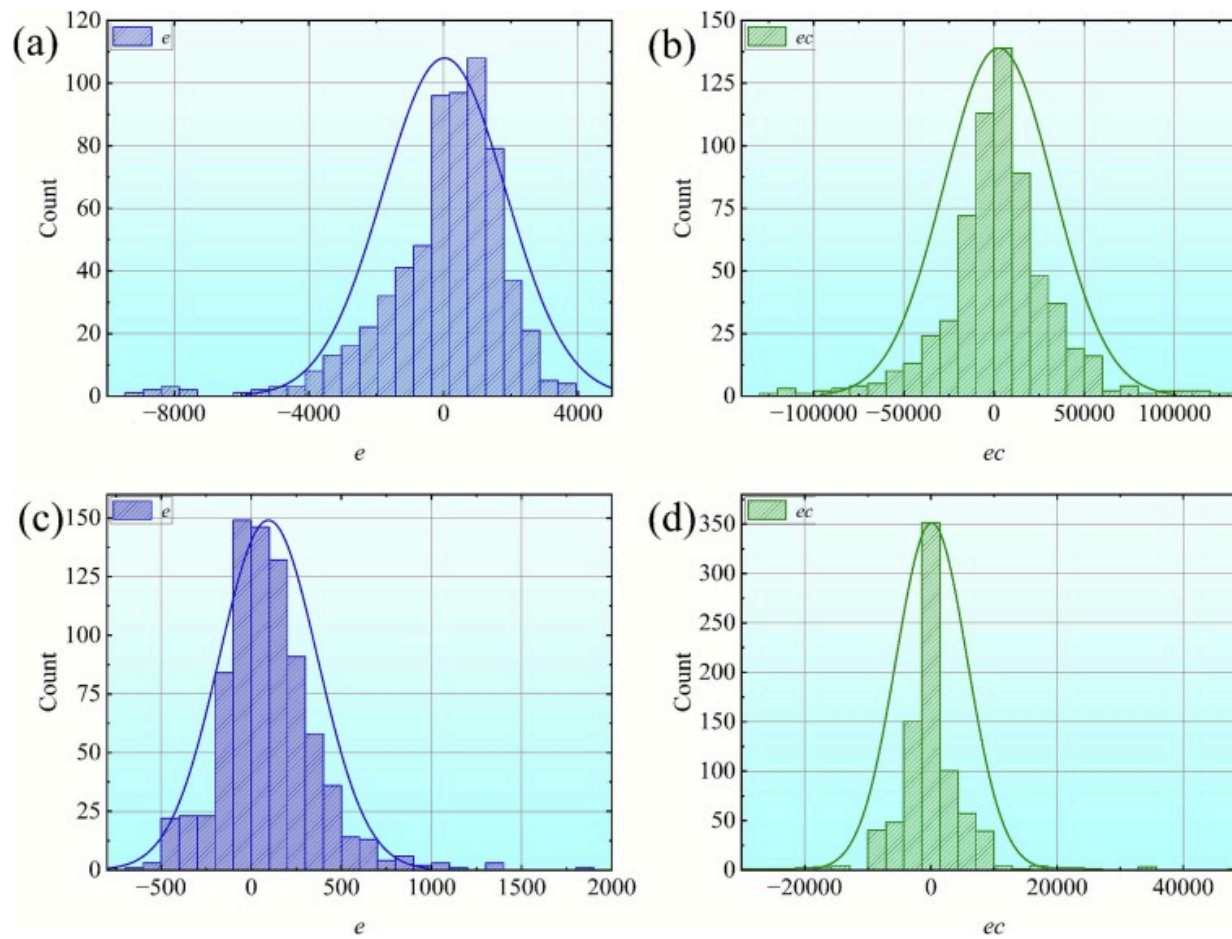
Table 5. Processing parameters of process database.

Experiment/Controller No.	Initial power	Processing thickness	Reference power
#1	6600W	flat plate=10mm	6600W

Experiment/Controller No.	Initial power	Processing thickness	Reference power
#2	5600W	gradient plate=8–10mm	5600/5400/5200W
	5600W	stepped plate=10mm	5600W
		stepped plate=9mm	5400W
		stepped plate=8mm	5200W

The reference power refers to the processing power used to generate the process database. It represents the power level under which consistent penetration depth can be achieved at different plate thickness positions. For flat-plate welding experiments, the reference power equals the initial power, meaning maintaining the initial power throughout the process uniform penetration. For variable-thickness plate welding, changes in plate thickness induce defocusing effects, necessitating dynamic power adjustments to maintain consistent penetration. The reference power represents the continuously adjusted power level during the welding process. When the control target is the stepped plate, experiments indicate that welding powers of 5600W, 5400W, and 5200W achieve consistent penetration depth for plate thicknesses of 10mm, 9mm, and 8mm, respectively. Therefore, these three power values are used as the reference power. When the control target is the gradient plate (with thickness continuously changing between 8–10mm), the reference power is determined based on the welding power required to achieve stable penetration depth under the conditions of maximum thickness, minimum thickness, and their average thickness, which is also set to 5600/5400/5200W.

Based on the reference power set by the process database, multiple spectral acquisition experiments were conducted under constant laser power welding of aluminum alloy. A total of 644 data sets were obtained for Controller #1, and 815 data sets for Controller #2. The results are shown in Fig. 8, where (a) and (b) represent the distributions of error e and rate of error change ec in the process database of Controller #1, while (c) and (d) show the corresponding distributions for Controller #2.



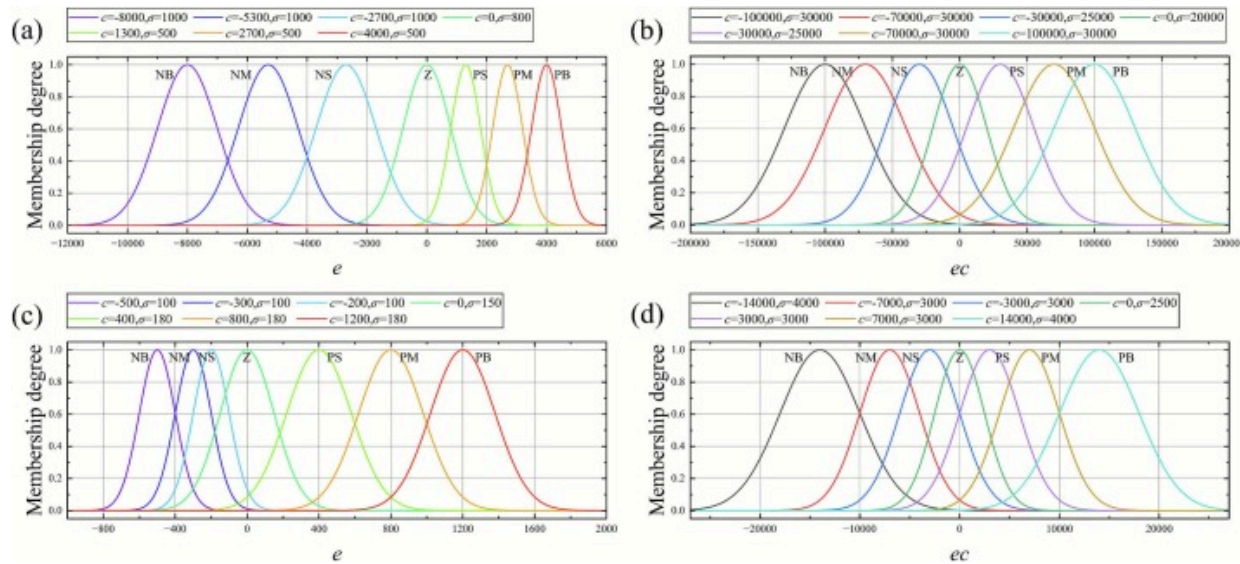
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Fig. 8. The data distribution of the process database. (a) The e of Controller #1; (b) The ec of Controller #1; (c) The e of Controller #2; (d) The ec of Controller #2.

As shown in Fig. 8, both error e and rate of error change ec exhibit a distribution pattern with higher density at the center and lower density at both ends, approximating a normal distribution but with a certain degree of asymmetry. Based on fuzzy logic, e and ec are each divided into seven linguistic variables: Negative Big (NB), Negative Medium (NM),

Negative Small (NS), Zero (Z), Positive Small (PS), Positive Medium (PM), and Positive Big (PB). These sections describe how much the input variable deviates from the reference value. Considering the actual distribution characteristics, the corresponding fuzzy membership functions are defined as shown in Fig. 9. The size of the set center c and width σ has been marked on the diagram.



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Fig. 9. The fuzzy membership function. (a) The e of Controller #1; (b) The ec of Controller #1; (c) The e of Controller #2; (d) The ec of Controller #2.

Based on the distribution characteristics of fuzzy membership function and to ensure that the controller maintains sufficient adaptability and working range under different process parameter conditions, the quantization domain of e and ec are set as $[-1200, 6000]$ and $[-200000, 200000]$, respectively for Controller #1; For Controller #2, the quantization domain of e and ec are set as $[-1000, 2000]$ and $[-27000, 27000]$ respectively for Controller #2. On this basis, the fuzzification layer of the FNN is constructed to ensure

that the controller can integrate prior process knowledge from experimental data, enabling adaptive fuzzy inference based on empirical data.

4.2. Implementation of controller

In the initial stage of feedback control, the PD controller is used to regulate penetration depth stability. The control law of the PD controller can be defined as:

$$P_{PD} = K_P e + K_D \dot{e} \quad (16)$$

where K_P is the proportional gain, and K_D is the derivative gain.

The electron temperature $T_{e_{stan}}$, which ensures stable penetration depth welding across multiple experiments, is taken as the reference value. During the welding process of Experiment, the real-time collected T_e signal is compared with $T_{e_{stan}}$ to obtain the error e and rate of error change $\dot{e}$, which serve as the controller inputs. The output power is calculated to modulate the welding power based on Eq. (20), ensuring penetration depth stability. After experimental validation, the values of $T_{e_{stan}}$ and the optimal K_P and K_D are shown in Table 6.

Table 6. Reference electronic temperature and PD controller parameter.

Experiment/ Controller No.	$T_{e_{stan}}$ (K)	K_P	K_D
#1	10,650	0.1	0.004
#2	8554	0.12	0.004

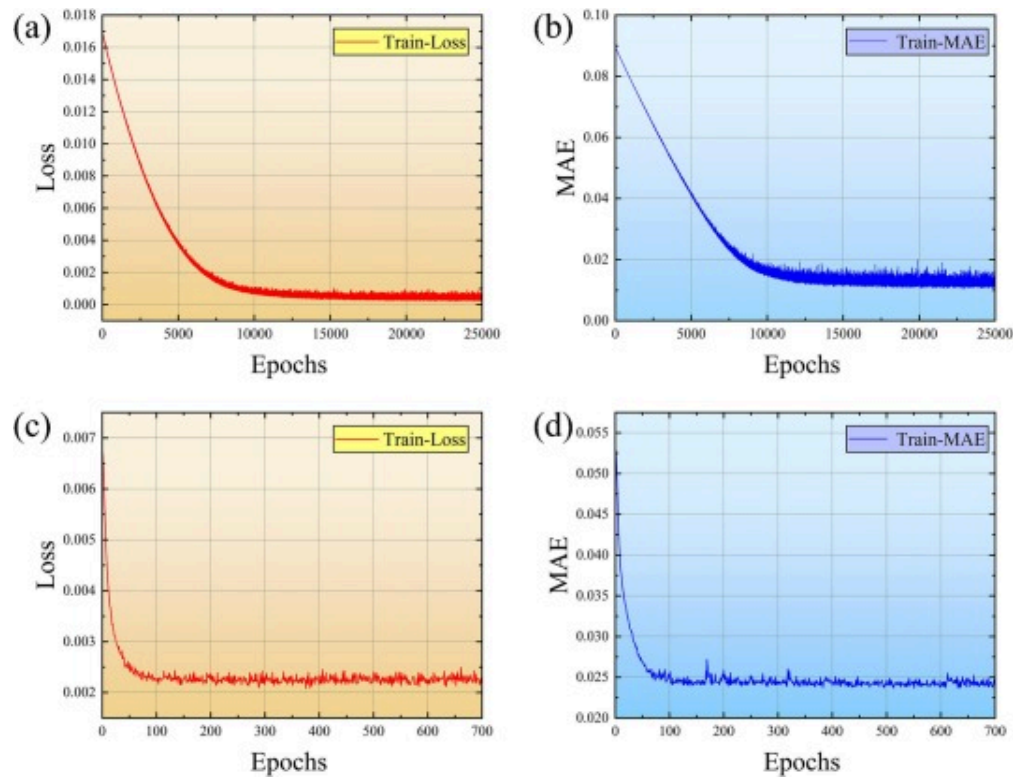
During the real-time feedback control phase of the PD controller, the system continuously records key variables, including the controller inputs e and $\dot{e}$, as well as the controller output P_{PD} . These data form the training dataset for the FNN controller, enabling it to learn and achieve control performance at least comparable to the PD controller. In this study, Controller #1 collected a total of 1,287 sample data points, while Controller #2

collected 2,446 sample data points. The FNN which has already incorporated prior process knowledge, undergoes online learning to further optimize control performance. The various hyperparameter settings for the FNN training process are shown in [Table 7](#).

Table 7. Hyperparameter setting during of FNN training.

Controller No.	Optimizer	Loss	Metrics	Epochs	Batch size
#1	Adam(learning rate=0.001)	MSE	MAE	25,000	256
#2	SGD(learning rate=0.00005, momentum=0.95)	MSE	MAE	700	256

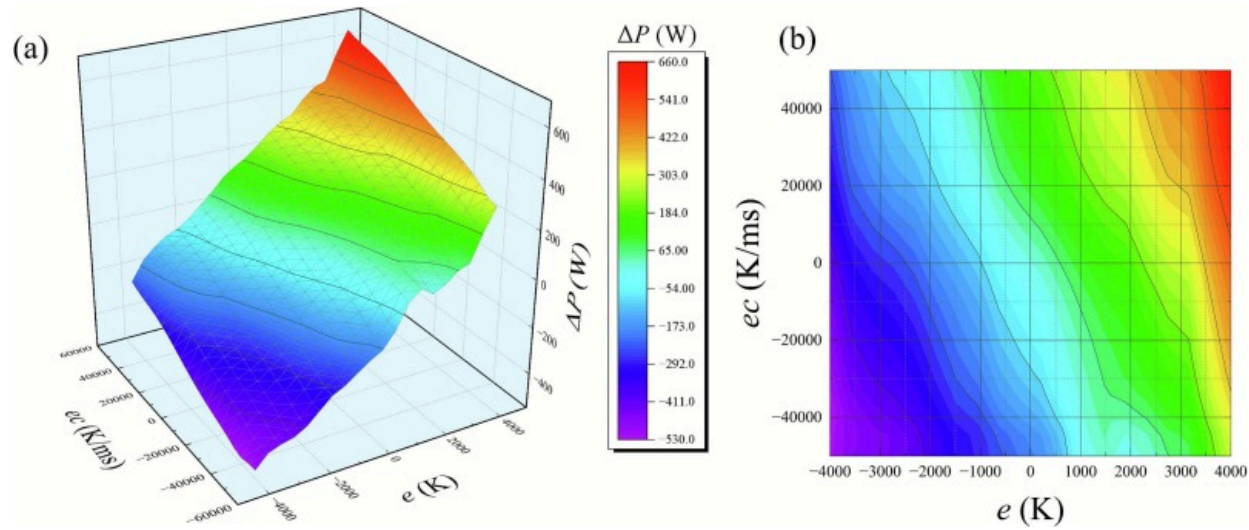
The learning records of Controller #1 and Controller #2 based on the FNN model are shown in [Fig. 10](#). Where (a) and (b) represent the loss values and accuracy, respectively, after normalization during the training process of Controller #1, while (c) and (d) correspond to Controller #2. It can be observed that the FNN of Controller #1 exhibits a training loss that decreases steadily in the early stages and converges around the 12,000th epoch, ultimately achieving a MAE of 0.01195. In contrast, the FNN of Controller #2 stabilizes much earlier, converging around the 100th epoch, and remains steady until the end of training, with a final MAE of 0.02501. Both controllers accurately learned the dynamic response characteristics of the PD controller in the welding system and effectively captured the nonlinear mapping relationship between power modulation and monitored signals during the welding process. The output response in the main operating range of the controllers are shown in [Figs. 11](#) and [12](#). The figures present the three-dimensional response surface of the modulated power along with its two-dimensional contour map.



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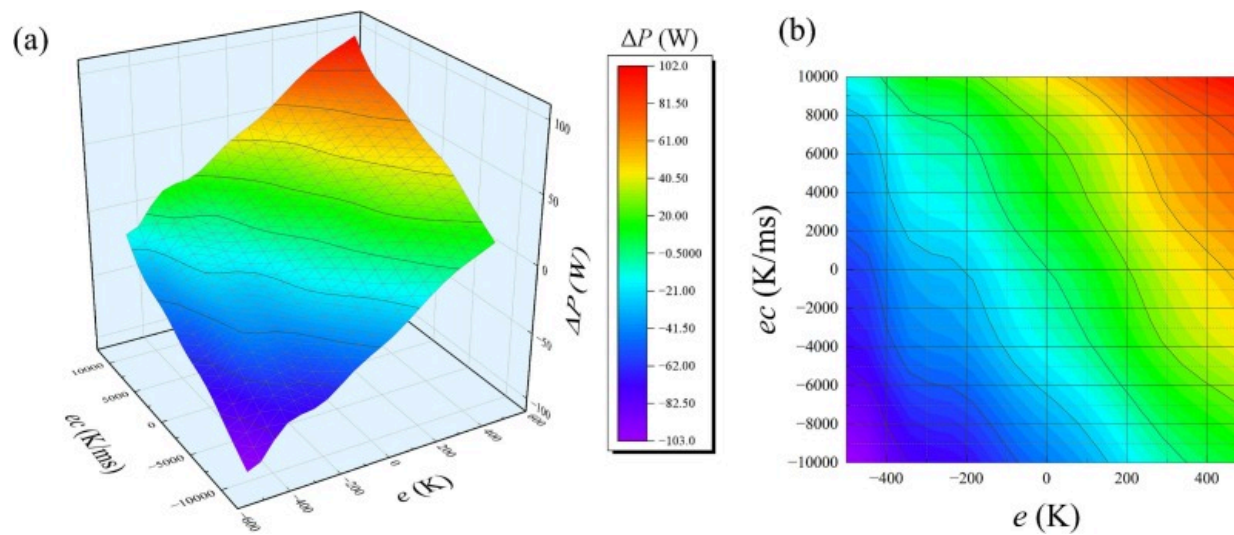
Fig. 10. FNN training history record. (a) Loss values of Controller #1; (b) MAE of Controller #1; (c) Loss values of Controller #2; (d) MAE of Controller #2.



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Fig. 11. The output results of Controller #1 (a) Three-dimensional response surface; (b) Two-dimensional contour map.



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Fig. 12. The output results of Controller #2 (a) Three-dimensional response surface; (b) Two-dimensional contour map.

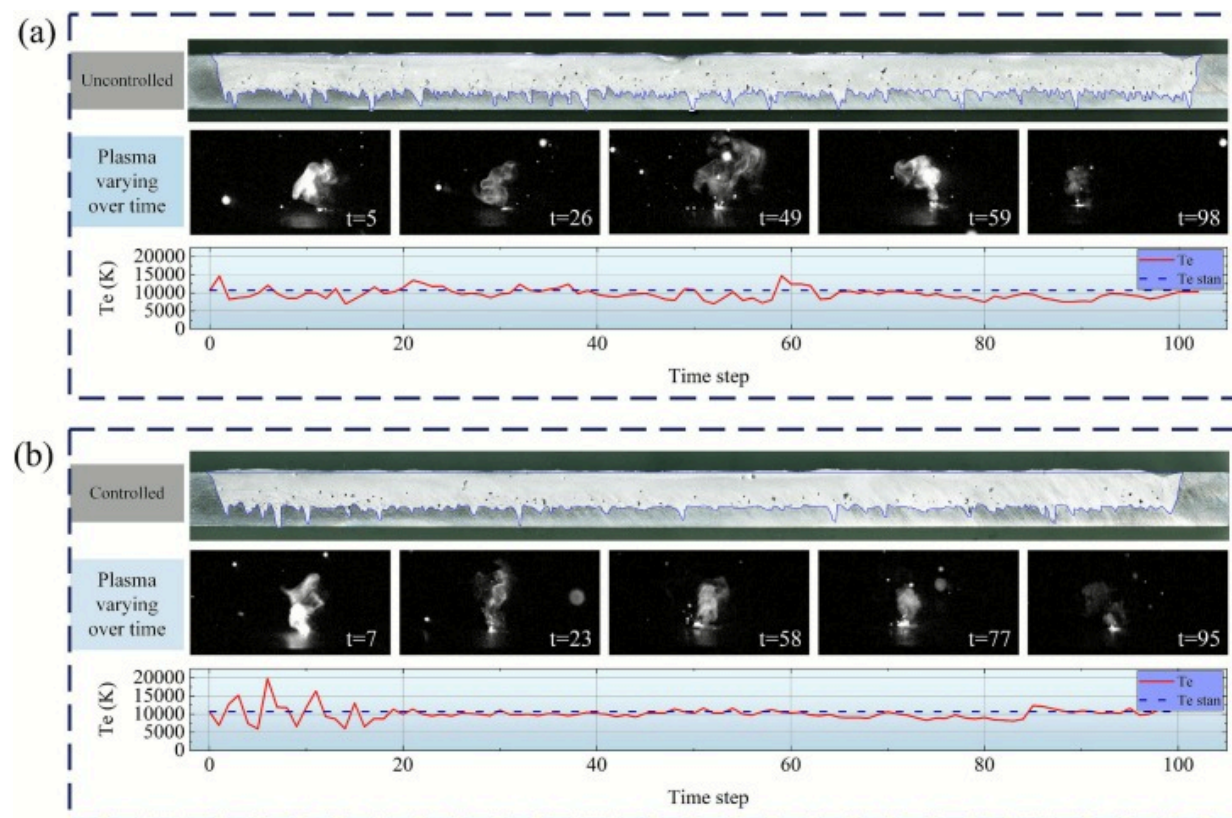
From Fig. 11, it can be observed that the modulated power of Controller #1 is mainly distributed within the range of $[-530, 660]$, indicating that during the feedback control process, the actual output power is adjusted around the reference power of 6600W. In contrast, Fig. 12 shows that the modulated power range of Controller #2 is narrower, at $[-104, 102]$. This is due to the different power adjustment strategy compared to Controller #1, as it adopts the cumulative update mode based on Eq. (19). After each feedback control time step, the power adjustment is incrementally updated with respect to the previous output value. In this mode, the modulation amplitude at each step remains small, thereby ensuring the stability of the controlled output power.

Additionally, the Figs. 11 and 12 demonstrate that the FNN effectively captures the overall trend of the modulated power response surface by learning from the PD controller data. Meanwhile, specific variations in power, such as local contour changes and non-uniform distributions, are further refined through the fuzzy layer settings, which embed prior process knowledge. Moreover, both e and ec show a clear positive correlation with the modulated power, consistent with the dynamic fluctuations in electron temperature and the actual welding conditions. These results validate the effectiveness and robustness of the FNN controller across different operating conditions.

4.3. Validation and comparison of control effects

The trained controller was validated by switching the output of the feedback control system to the FNN model for real-time feedback control in aluminum alloy laser welding. The control effect for flat plate welding in Controller #1 is shown in Fig. 13, which includes the longitudinal cross-section of the weld after etching, the plasma fluctuation over time, and the monitored electron temperature signal. The Fig. 13 indicates that the controlled weld penetration stability is significantly better than the uncontrolled state.

From the captured plasma images, it can be observed that in the initial welding stage, the interaction between the laser and the aluminum alloy has not yet stabilized, leading to large energy absorption fluctuations and more intense plasma fluctuations compared to the steady-state condition. As welding progresses, sporadic oscillations of the keyhole and molten pool in the uncontrolled case continue to intensify plasma ejections, causing penetration fluctuations, as seen at $t=49$ and $t=59$ in Fig. 13(a). However, in Fig. 13(b), after control is applied, the controller adjusts the laser energy output in response to detected plasma fluctuations, thereby suppressing keyhole and molten pool instabilities and ensuring continuous penetration stability.

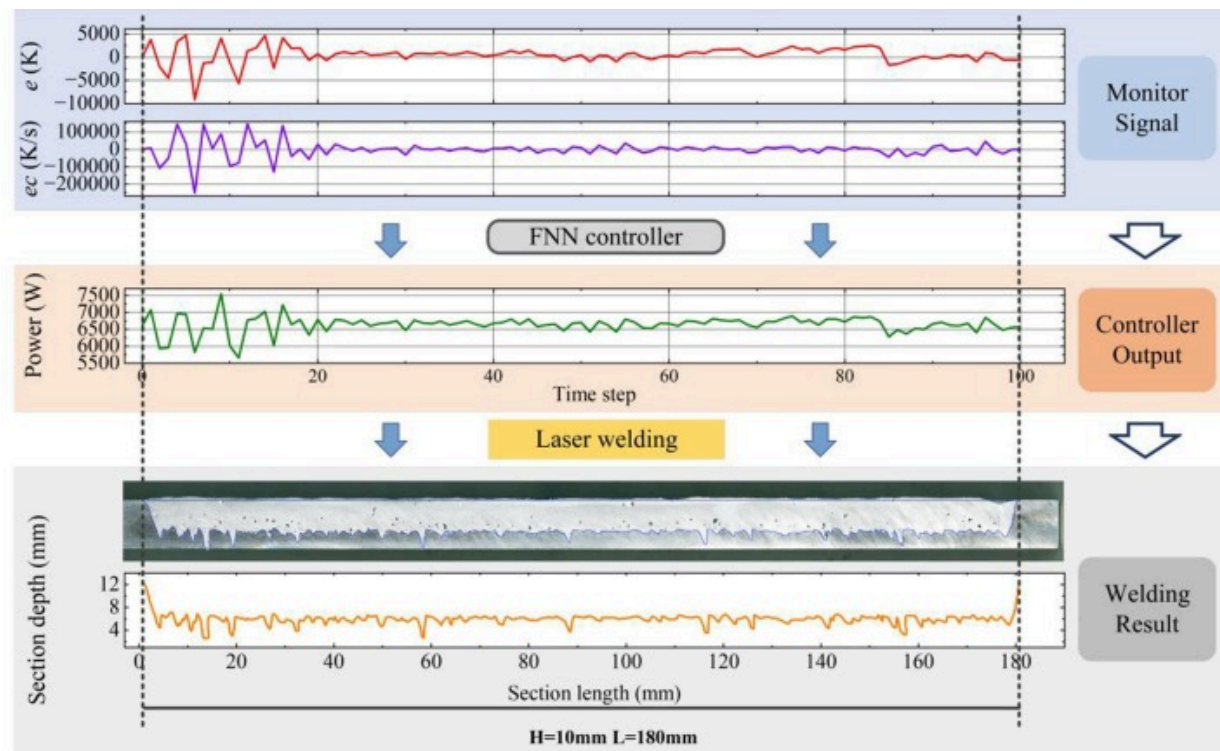


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Fig. 13. Comparison of control effects in flat plate welding (a) Uncontrolled welding process; (b) Controlled welding process.

Fig. 14 illustrates the variations in electron temperature and real-time power output during the control process. At the initial stage of welding, the power exhibits pronounced fluctuations, which are likely attributable to control overshoot induced by strong signal variations. As welding proceeds, the monitored e and ec remain within a certain range, indicating that no severe fluctuations occur in the welding process. Meanwhile, the output power remains around the baseline power of 6600W, avoiding excessive power adjustments that could lead to penetration instability.

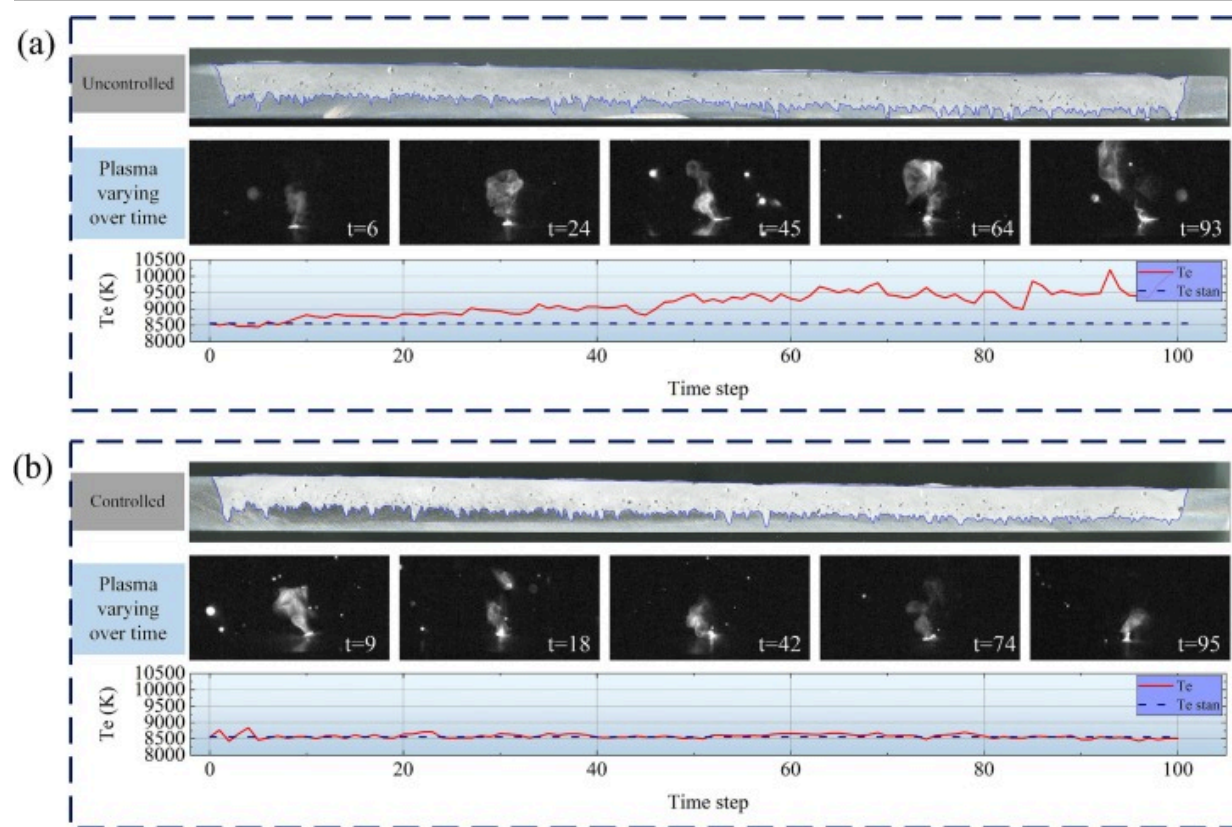


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Fig. 14. Flat plate welding control process record.

The effect of penetration fluctuation control using Controller #2 for the gradient plate is shown in Fig. 15. During the welding process, the middle section of the plate continuously decreases in height, causing the defocus amount to gradually reduce from -3 mm to -1 mm. As seen in Fig. 15(a), as the defocus amount decreases, the energy input becomes more concentrated. Without control, the penetration depth continuously increases, and partial burn-through occurs at the weld end. Meanwhile, the plasma ejection intensifies over time, and the monitored electron temperature signal rises accordingly. In contrast, Fig. 15(b) shows that with control applied, despite the varying defocus amount, both penetration depth and the collected plasma signal remain stable, demonstrating the effectiveness of the controller.

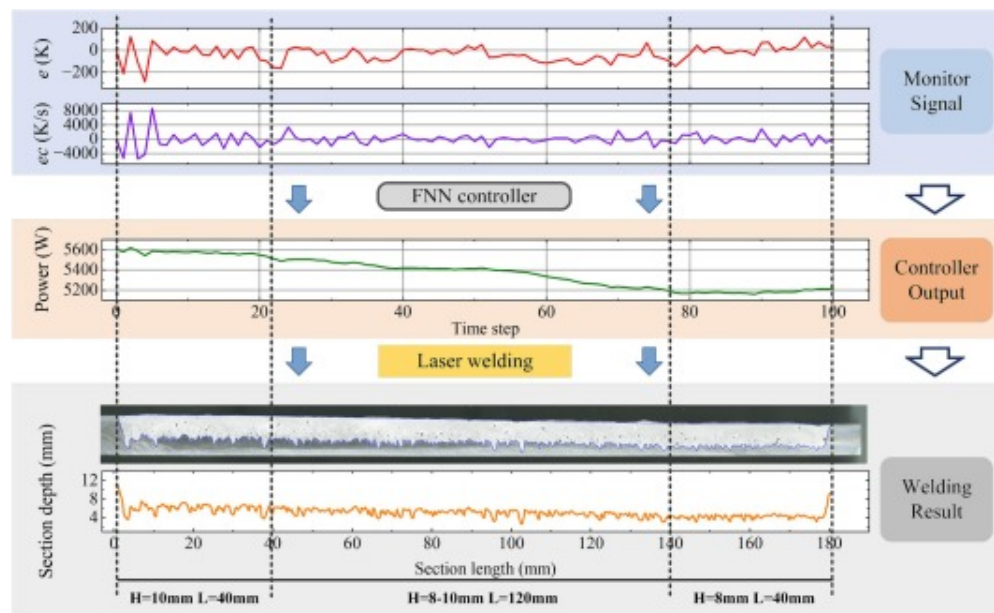


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Fig. 15. Comparison of control effects in gradient plate welding (a) Uncontrolled welding process; (b) Controlled welding process.

Fig. 16 presents the control process record for gradient plate welding. During the 120mm middle section where the defocus amount changes, the collected electron temperature error signal is mostly within the range of $[-100, 0]$, indicating a trend of increasing penetration depth. In response, the output power continuously decreases to maintain penetration stability. In the 40mm sections at both the front and rear of the weld, where the defocus amount remains unchanged, the output power remains stable as well.



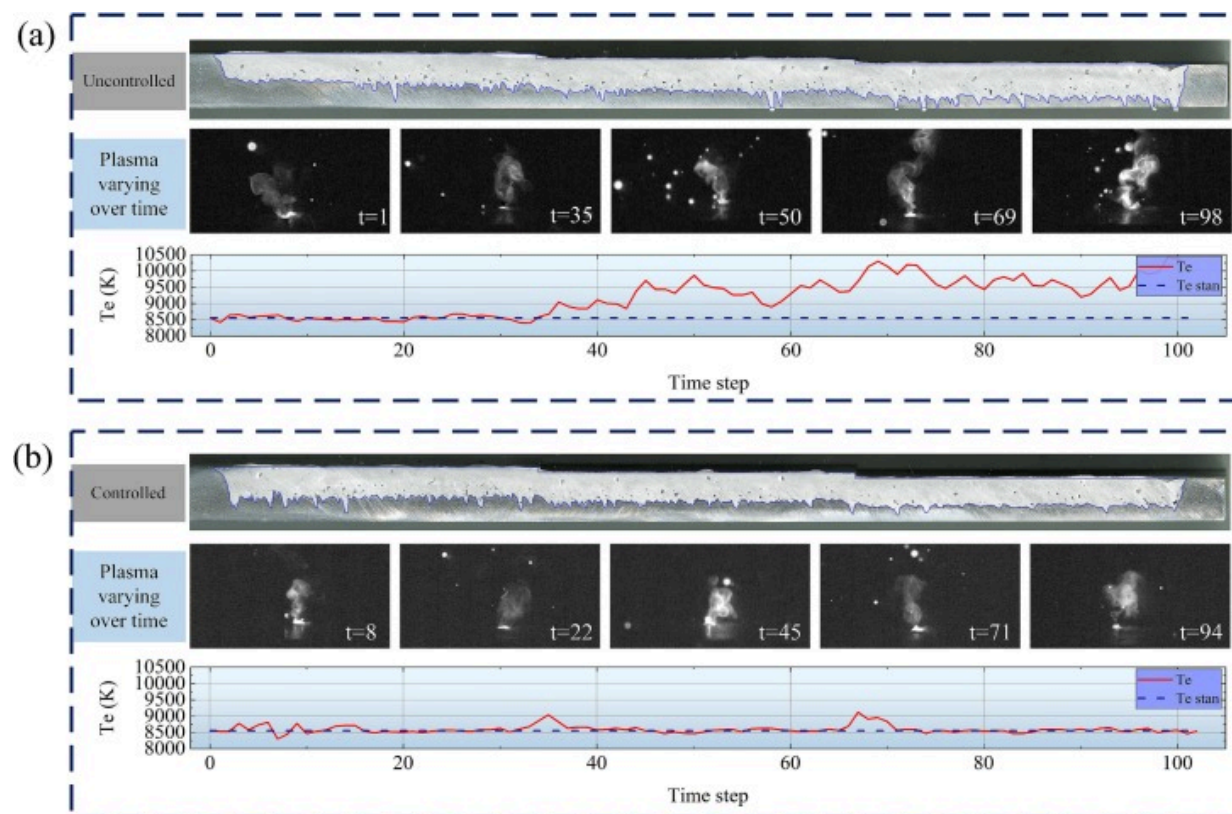
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Fig. 16. Gradient plate welding control process record.

The effect of penetration fluctuation control using Controller #2 for the stepped plate is shown in Fig. 17. During the welding process, the plate thickness undergoes two abrupt changes, decreasing from 10mm to 9mm and then further reducing to 8mm. As a result,

without control, the variation in defocus causes two noticeable increases in penetration depth, as illustrated in Fig. 17(a). Under the same defocus condition, the penetration remains uniform within each thickness region; but significant differences in penetration depth are observed among the three thicknesses. At the end of welding, partial burn-through also occurs. Meanwhile, two distinct increases in plasma intensity and electron temperature are observed. Specifically, plasma concentration begins to rise at $t=35$ and $t=50$, followed by pronounced plasma ejection at $t=69$ and $t=98$. In contrast, Fig. 17(b) shows that with control applied, welding remains stable despite the variations in defocus, and the monitored signals do not exhibit severe fluctuations.

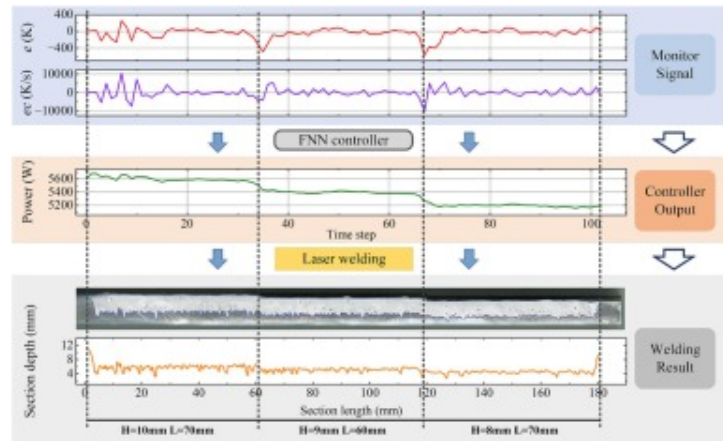


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Fig. 17. Comparison of control effects in stepped plate welding (a) Uncontrolled welding process; (b) Controlled welding process.

The control process record for stepped plate welding is shown in Fig. 18. The monitoring signals clearly indicate significant fluctuations in electron temperature e and ec at the two points where the height change causes abrupt defocus variations, corresponding to a sudden increase in energy input. Meanwhile, the controlled power is adjusted within 2–3 times steps to stabilize, preventing further penetration depth increase. The results demonstrate that penetration remains consistently stable. Additionally, the stabilized output power values closely match the power levels identified in the process database for maintaining stable penetration, further validating the accuracy of the power modulation.



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Fig. 18. Stepped plate welding control process record.

To further quantify the penetration stability under this feedback control system, this study employs the Coefficient of Variation (CV) of penetration depth to measure fluctuation magnitude. The specific calculation method is as follows:

$$CV = \frac{\sigma_d}{\bar{d}} \times 100\% \quad (17)$$

where σ_d is the standard deviation of penetration depth, representing the fluctuation magnitude of penetration data, and $\bar{d}$ is the mean penetration depth, indicating the overall level of standard penetration.

$$\sigma_d = \sqrt{\frac{1}{N} \sum_{i=1}^N (d_i - \bar{d})^2} \quad (18)$$

$$\bar{d} = \frac{1}{N} \sum_{i=1}^N d_i \quad (19)$$

In this study, $\bar{d}$ is chosen as the mean value of the tested stable target penetration depth, which is used to evaluate the deviation and stability of the weld penetration depth relative to the target value.

The quantified results of penetration depth fluctuation are shown in Table 8. It can be observed that the *CV* of penetration depth fluctuation is significantly reduced after applying the control, further validating the effectiveness of the control system.

Table 8. The CV of penetration depth under different experiments.

Workpiece	Flat plate (10mm)	Gradient plate (8–10mm)	Stepped plate (8–10mm)
Uncontrolled CV	18.84%	17.17%	19.77%
Controlled CV	11.08%	10.25%	12.07%

5. Conclusions

This study proposes a parallel self-learning FNN feedback control method based on the relationship between penetration depth and spectral signals in aluminum alloy laser welding. By integrating a process database and experience-based strategy learning of the PD controller, this approach effectively stabilizes penetration depth fluctuations in both flat and variable-thickness plate welding. The main conclusions are summarized as follows:

(1) The real-time monitored electron temperature signal accurately reflects penetration depth fluctuations. In flat plate welding, severe penetration depth variations caused by the intense evaporation of Mg or other elements are manifested as sudden changes in the electron temperature signal. In variable-thickness plate welding, as the defocus amount decreases and penetration depth increases, the electron temperature also exhibits an upward trend.

(2) During training of the controller, the two networks responsible for controlling flat and variable-thickness plate achieve final MAE values of 0.01195 and 0.02501, respectively. The proposed structure effectively learns the dynamic characteristics of the PD controller within the welding control system, while simultaneously integrating prior process knowledge, thereby ensuring precise power modulation output.

(3) The parallel self-learning FNN controller significantly reduces sporadic penetration depth fluctuations during flat plate welding, lowering the penetration depth CV to 11.08%. In gradient plate welding, as the defocus amount gradually decreases, the controller smoothly reduces power to ensure the penetration depth consistency, with the CV decreasing to 10.25%. In stepped plate welding, the controller promptly adjusts power in response to sudden defocus changes and ensures stable penetration depth across three thicknesses, with the CV reduced to 12.07%.

To further improve the accuracy and response speed of the control system, future efforts will focus on the following aspects: employing multi-sensor monitoring with cameras and spectrometers, integrating deep learning models for a more precise representation of penetration depth fluctuations in the welding process, and optimizing the computational efficiency of the control algorithm to enhance real-time performance.

CRedit authorship contribution statement

Yuan Wang: Writing – original draft, Methodology, Investigation, Conceptualization.

Bangyan Xu: Resources. **Congyi Wu:** Software. **Ping Jiang:** Project administration. **Chu**

Han: Supervision. **Leshi Shu:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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